

Project Test Execution Report

1. Test Execution Summary

A complete validation sweep was executed on the HDI Predictor platform. Testing covered Unit tests (for inputs, thresholds, and database writes), Integration tests (for Flask routing endpoints), System tests (for end-to-end user flows), and User Acceptance Testing (UAT).

Result: All 20 core test cases passed successfully. No memory leaks or syntax errors were identified.

2. Test Coverage & Breakdown

- Unit Tests: 8 cases (Input boundary checking, scaler transforms, database schema checks).
- Integration Tests: 6 cases (Route /predict status, /compare json response, PDF download generation).
- System / UAT Tests: 6 cases (Theme switching persistence, dynamic gauge rendering, CSV logs output).